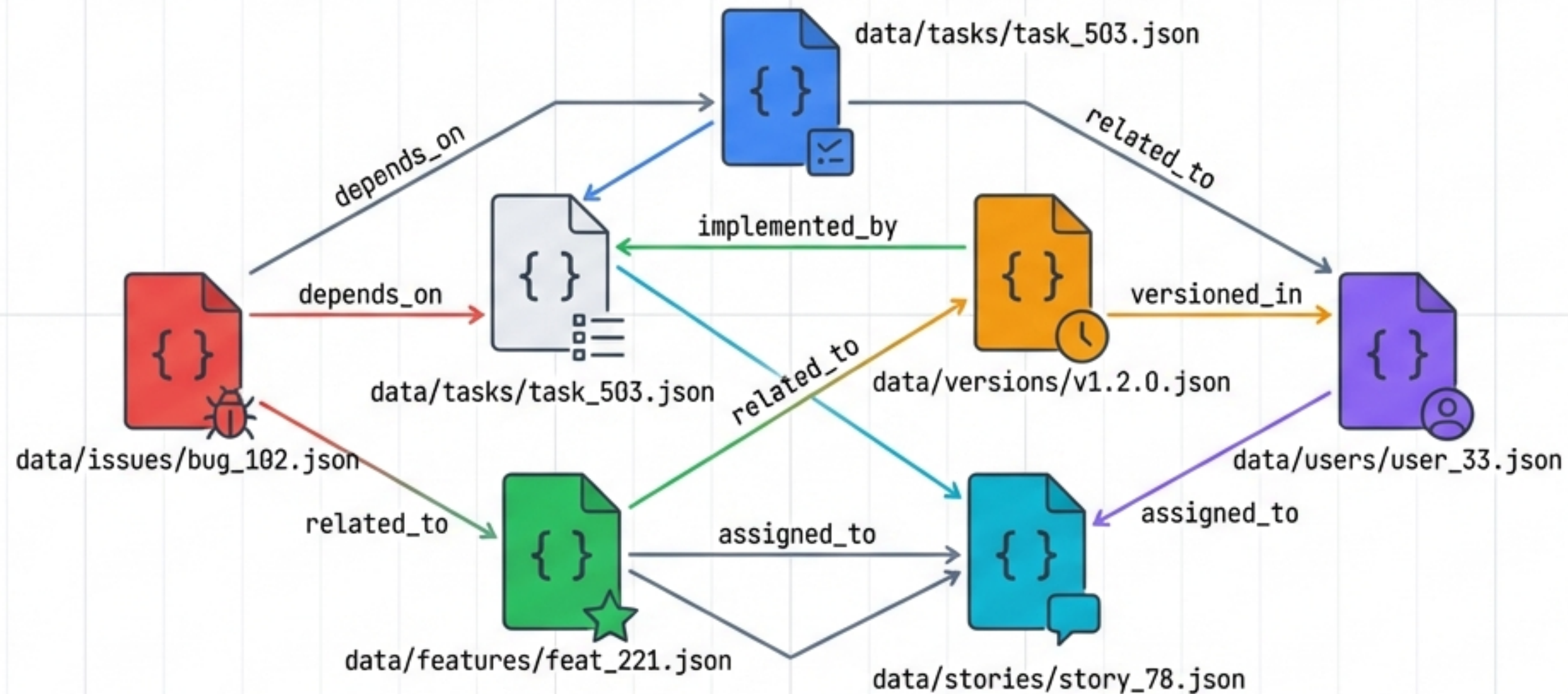


Git-Native Issue Tracking System

BacBrend Development Briefing // v0.2.19



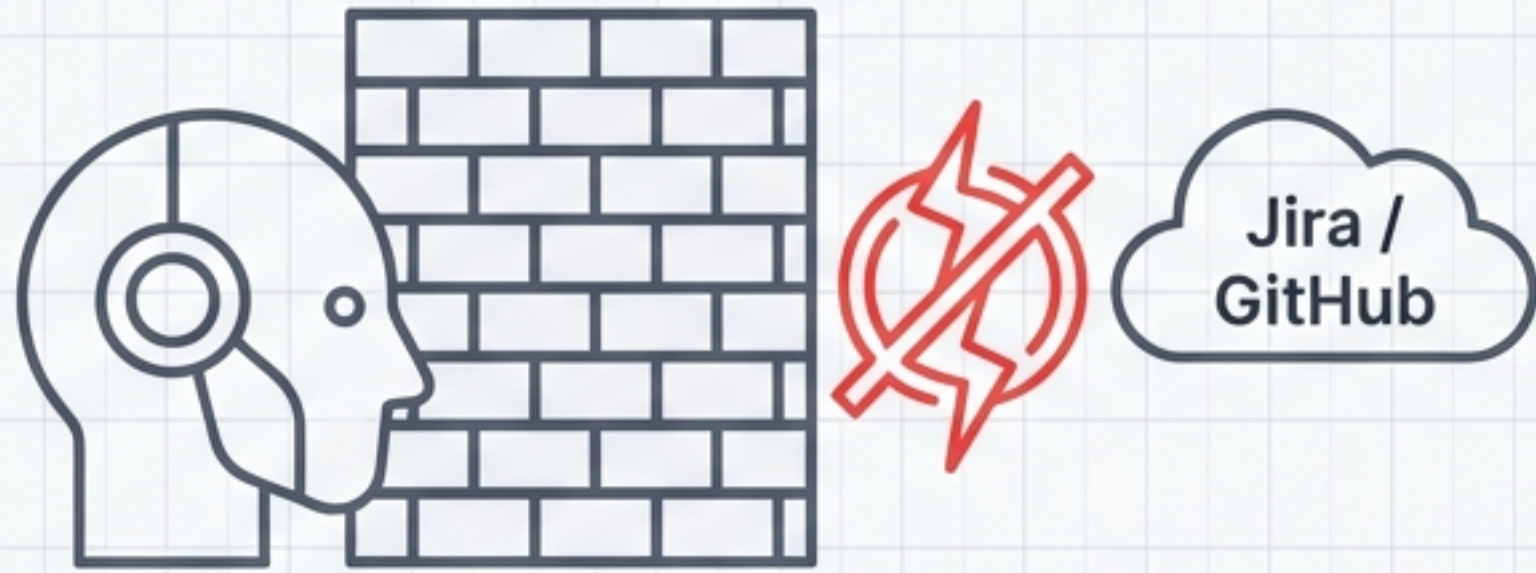
LAST UPDATED
2026-02-01

SCOPE
Backend/API & Coordination

TARGET
New Claude Session

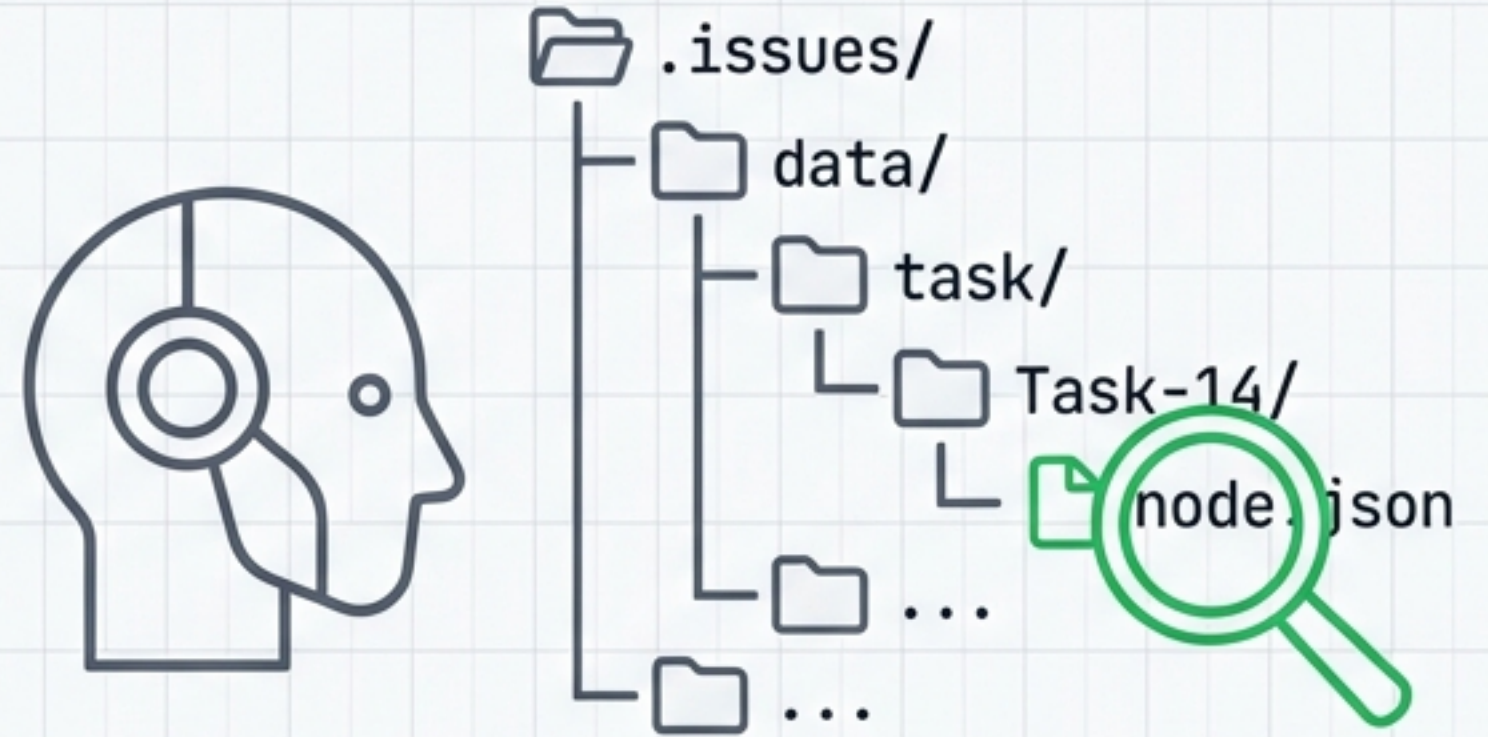
The Problem: The AI Blindness of External Tools

The Friction



Traditional trackers live outside the repository.
When you tell Claude Code to “implement the backlog,” it cannot see it.

The Solution



Core Philosophy: Issues **are** the codebase.
Mechanism: Stored as JSON files in ``.issues/``.
Result: Direct read access to ``.issues/data/task/Task-14/node.json``.

The Meta-Loop: Building the Tool With the Tool

We are using this issue tracking system to develop itself.



Note: Everything—code, issues, tasks, and bugs—is versioned together in Git.

System Architecture & The Graph

Stack:

FastAPI (Backend Focus)

Storage:

Filesystem-based
(No external database)

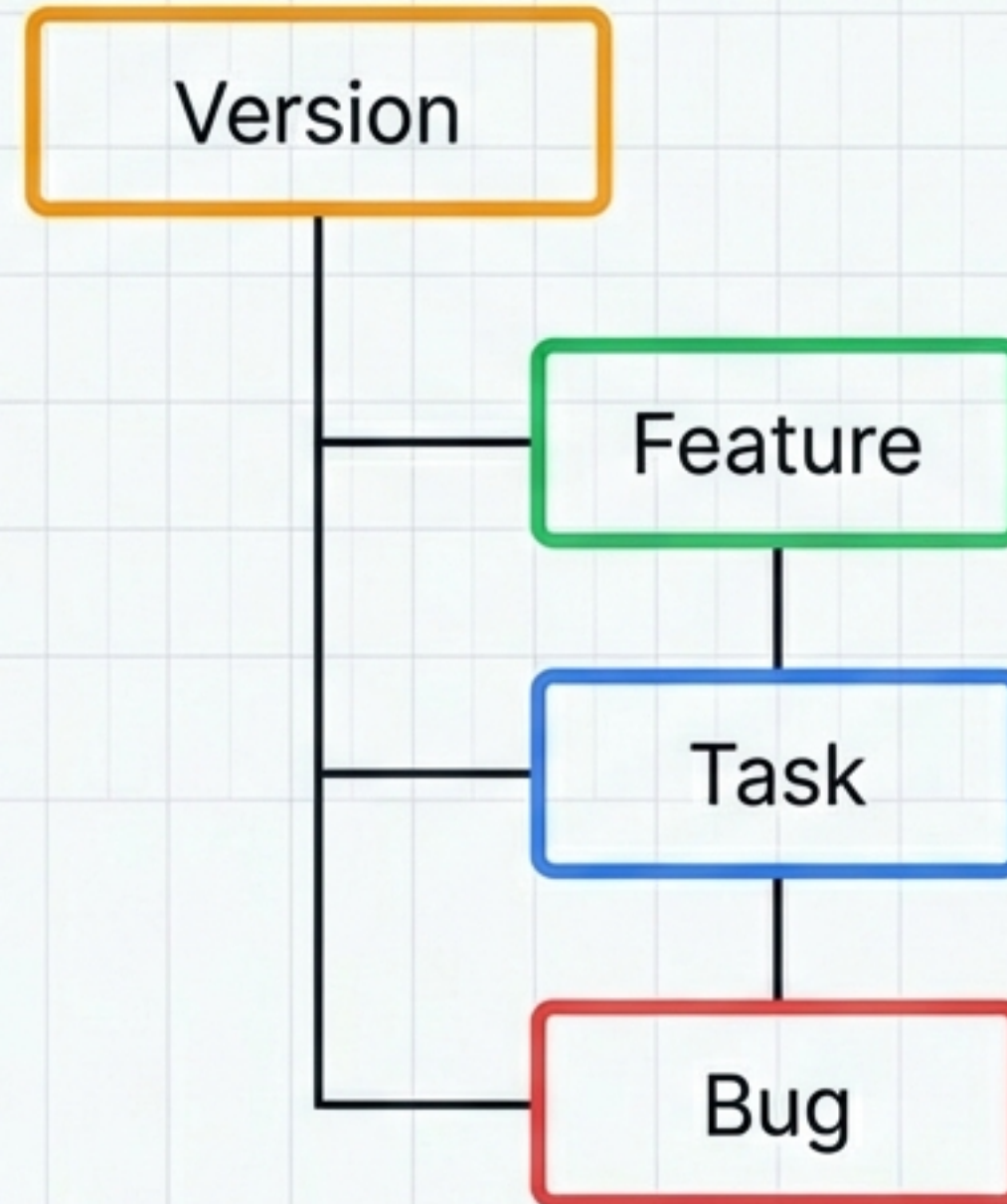
Data Structure:

A Node Graph

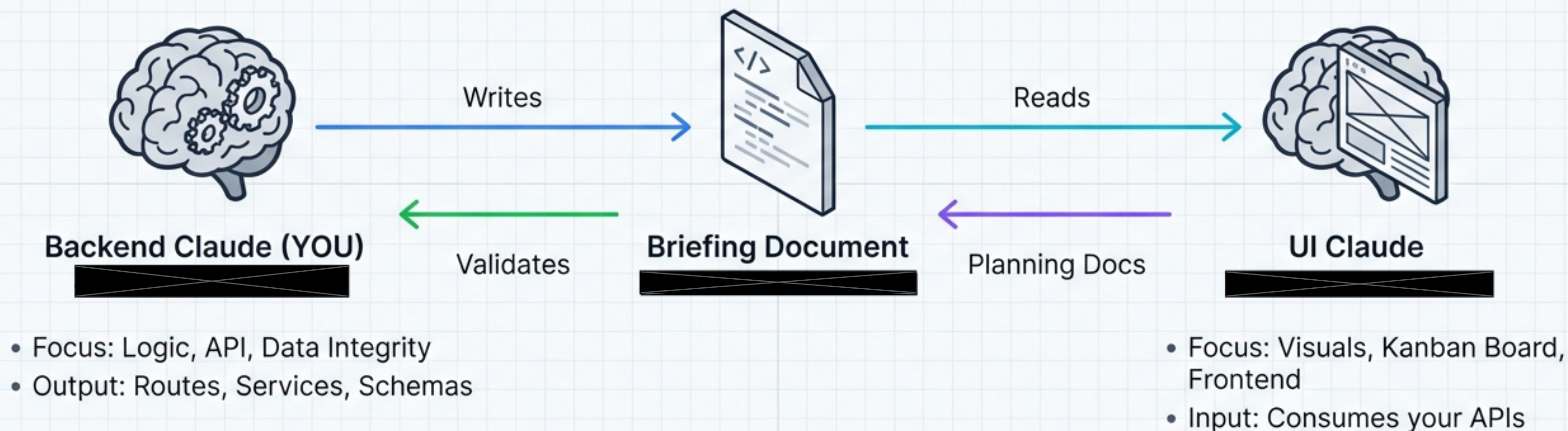
Query Method:

Directory traversal & JSON parsing

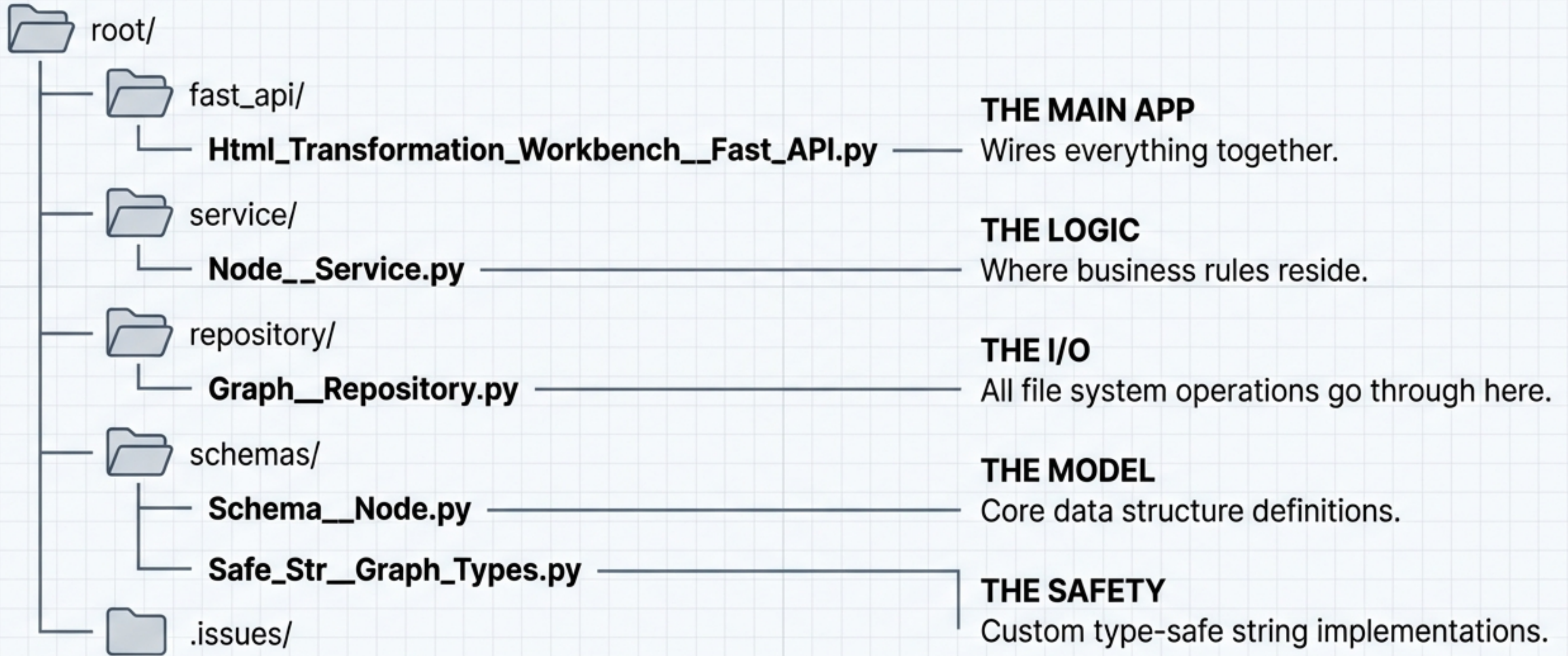
The Entity Graph



The Multi-Claude Pattern



Project Structure: The Backend Map



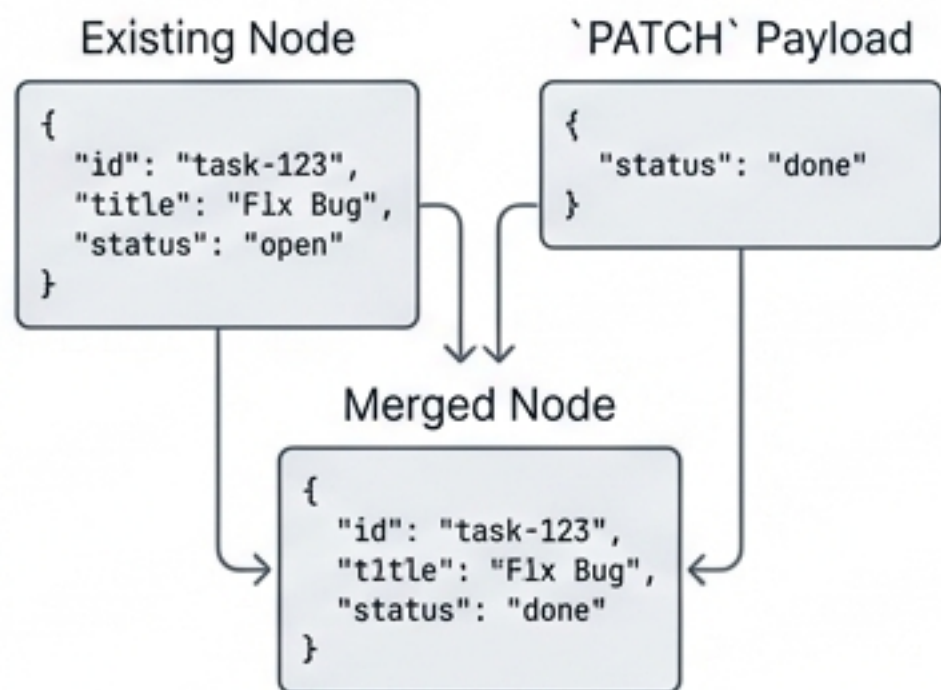
Your Mandate: The Backend Workflow



Key Pattern: The Service Layer

Business Logic Residence: ``Node__Service.py``

Deep Merge



``PATCH`` requests preserve existing properties. Does not overwrite the whole node.

Validation



Ensures graph links (e.g., Task → Feature) are valid before saving.

Safety



`Graph_Repository.py`

``Graph_Repository.py`` handles physical file I/O to prevent corruption.

The Data Model: Node Types & Statuses

BUG

Backlog, Confirmed,
In-Progress, Resolved

TASK

Backlog, Todo, Done

FEATURE

Proposed, Approved,
Released

VERSION

Planned, Deprecated

USER-STORY

Draft, Validated

PERSON

Active, Inactive

Coordination Protocol

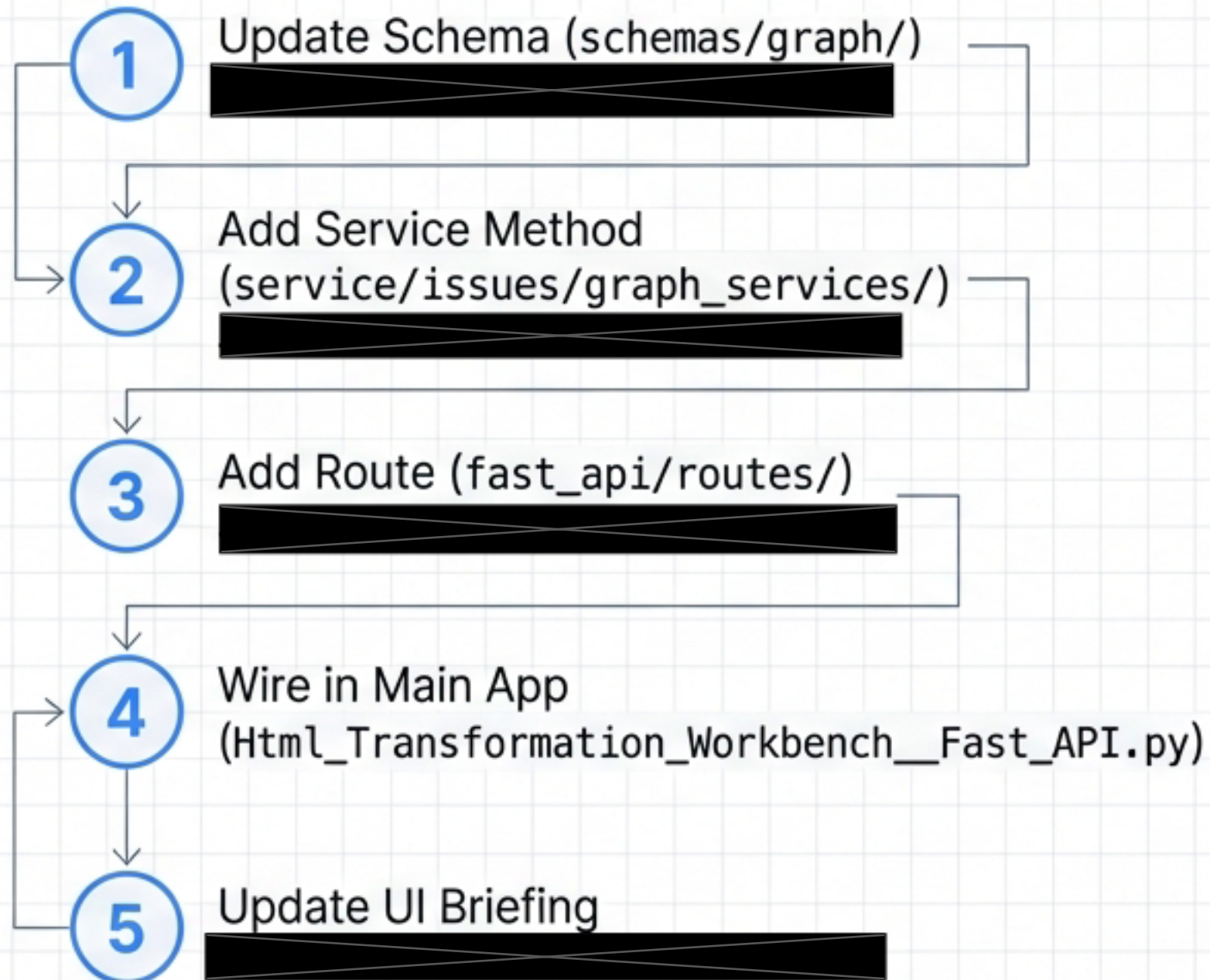
Managing the Handshake with UI Claude

TRIGGER: When to write a UI Briefing?

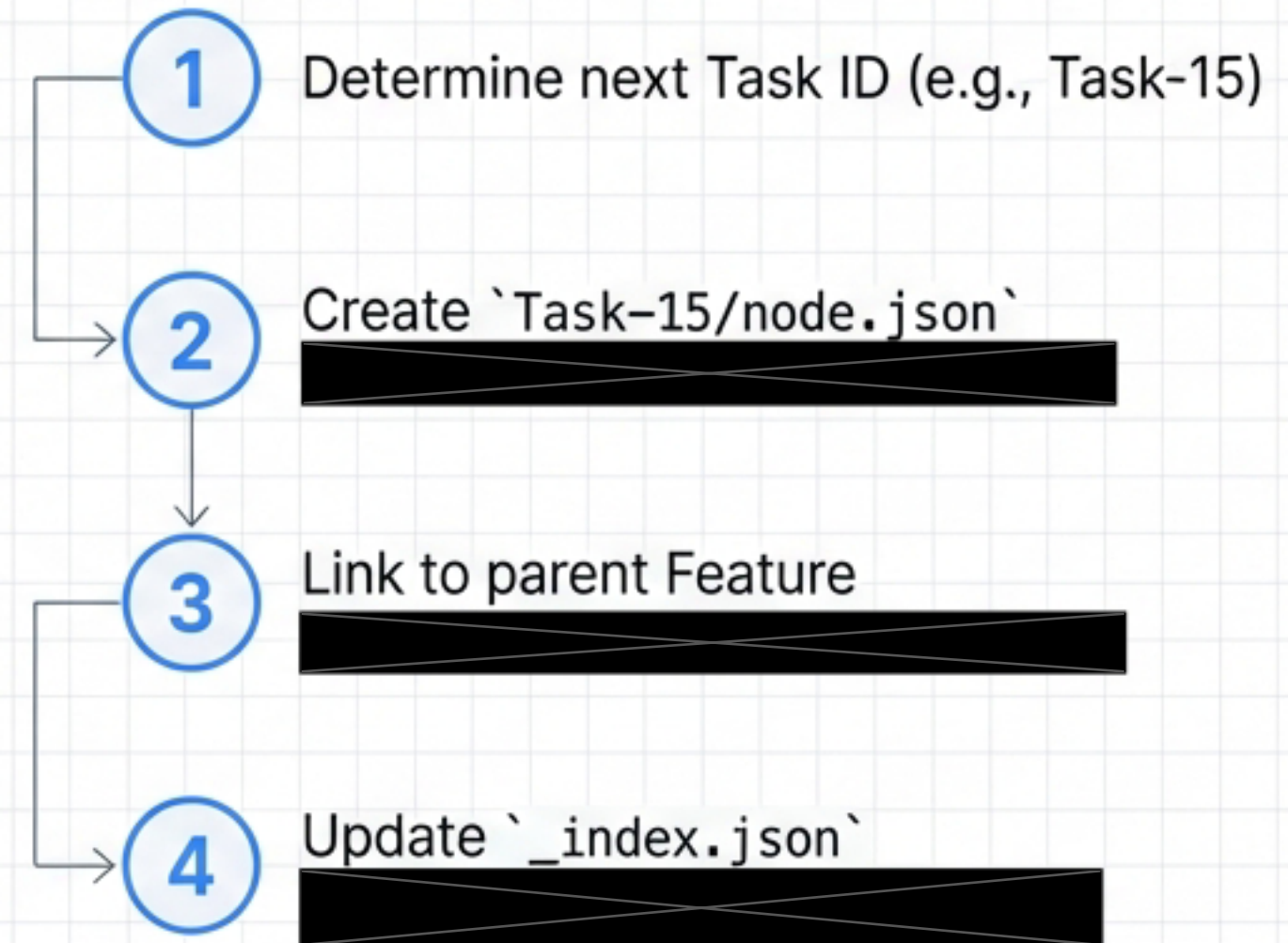


Common Tasks: Execution Guide

Scenario A: Adding an API Endpoint



Scenario B: Creating a Backlog Task



Current Status: v0.2.19

Completed Features

- ✓ Graph Traversal API: ``GET /nodes/api/nodes/{type}/{label}/graph``
- ✓ Deep Merge: ``PATCH`` works correctly
- ✓ Comments API: Dedicated CRUD endpoints
- ✓ Safety: Create Node prevents overwrites
- ✓ Multiline Descriptions: Newlines preserved

Backend is functional. Graph API is ready for UI integration.

Known Issues & Technical Debt

The Watch Out



The Watch Out

- Index Sync: `_index.json` can get out of sync; rely on safety checks.
- Type Init: Remember `'Safe_Str'` initializes to empty string, not `None`.

Refactoring Needed



Refactoring Needed

- `'Schema__Node.properties'` uses `'Dict[str, Any]'`—needs typing.
- Standardizing 'Issue' naming conventions.

Immediate Backlog



Immediate Backlog

- [Task-3](#): Comments UI support (**High Priority**)
- [Task-6](#): Graph visualization components

Start Sequence